

Statistical Modeling

DATA.STAT.740

Lecture 6

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Exponential family of distributions (EFD)

Likelihood $L(\theta_i, \phi) = \exp \left(\frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right)$

Log-likelihood $l(\theta_i, \phi) = \log(L(\theta_i, \phi)) = \frac{y_i \theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)$

The mean and the variance of these distributions are of the form

$$\mu_i = E(\mathbf{y}_i \mid x_i) = b'(\theta_i), \text{Var}(\mathbf{y}_i \mid x_i) = b''(\theta_i)a(\phi)$$

Canonical link function is defined by $g(\mu_i) = \theta_i$

Using canonical link means $\eta_i = x_i^T \beta = \theta_i, \quad \mu_i = b'(\eta_i)$

Canonical links

Link function $g(\mu) = \eta$

Canonical link $\eta = \theta$

Now
$$\frac{\partial \mu}{\partial \eta} = \frac{\partial \mu}{\partial \theta}$$

From before
$$\mu = b'(\theta) \Rightarrow \frac{\partial \mu}{\partial \theta} = b''(\theta)$$
$$\text{Var}(\mathbf{y}) = b''(\theta)a(\phi) \Rightarrow b''(\theta) = \frac{\text{Var}(\mathbf{y})}{a(\phi)}$$

Therefore for canonical links we have

$$\frac{\partial \mu}{\partial \eta} = \frac{\text{Var}(\mathbf{y})}{a(\phi)}$$

Remember maximum likelihood estimator?

The general formulation for GLMs is that we solve the score equations for β

$$\sum_{i=1}^N \frac{y_i - \mu_i}{\text{Var}(\mathbf{y}_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot x_i = 0$$

where the dependence of β is through $\mu_i = g^{-1}(x_i^T \beta)$

$$\mathbf{u}_\beta = \frac{\partial l(\beta \mid \mathbf{y})}{\partial \beta} = X^T D V^{-1}(\mathbf{y} - \mu)$$

Estimator $\hat{\beta}$ for which $\mathbf{u}_{\hat{\beta}} = X^T \hat{D} \hat{V}^{-1}(\mathbf{y} - \hat{\mu}) = 0$, is called **maximum likelihood estimator** of β

Remember maximum likelihood estimator?

The general formulation for GLMs is that we solve the score equations for β

$$\sum_{i=1}^N \frac{y_i - \mu_i}{\text{Var}(\mathbf{y}_i)} \cdot \frac{\partial \mu_i}{\partial \eta_i} \cdot x_i = 0$$

Let's plug in the canonical link equality $\frac{\partial \mu}{\partial \eta} = \frac{\text{Var}(\mathbf{y})}{a(\phi)}$

We get
$$\sum_{i=1}^N \frac{y_i - \mu_i}{a(\phi)} \cdot x_i = 0$$

Remember weighted least squares estimation?

Let $\hat{\beta}$ be the maximum likelihood **estimator** for the GLM $g(\mu) = \eta = X\beta$

Asymptotically we can approximate $\hat{\beta} \sim \text{Normal}(\beta, F_{\beta}^{-1})$

where

$$\begin{aligned} F_{\beta} &= \text{E} \left(X^T D V^{-1} (\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} \text{E} \left((\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} V V^{-1} D X = X^T W X \end{aligned}$$

Typo last week!

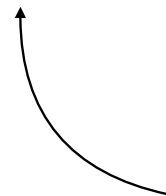
Plug in $\frac{\partial \mu}{\partial \eta} = \frac{\text{Var}(\mathbf{y})}{a(\phi)}$

$$W = D V^{-1} D = \begin{bmatrix} \frac{\left(\frac{\partial \mu_1}{\partial \eta_1}\right)^2}{\text{Var}(\mathbf{y}_1)} & 0 & \dots & 0 \\ 0 & \frac{\left(\frac{\partial \mu_2}{\partial \eta_2}\right)^2}{\text{Var}(\mathbf{y}_2)} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{\left(\frac{\partial \mu_N}{\partial \eta_N}\right)^2}{\text{Var}(\mathbf{y}_N)} \end{bmatrix}$$

Remember weighted least squares estimation?

Fisher information matrix becomes

$$\begin{aligned} F_{\beta} &= \mathbb{E} \left(X^T D V^{-1} (\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} \mathbb{E} \left((\mathbf{y} - \mu) (\mathbf{y} - \mu)^T V^{-1} D X \right) \\ &= X^T D V^{-1} V V^{-1} D X = X^T W X \end{aligned}$$


$$W = D V^{-1} D = \frac{1}{a(\phi)^2} \begin{bmatrix} \text{Var}(\mathbf{y}_1) & 0 & \cdots & 0 \\ 0 & \text{Var}(\mathbf{y}_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \text{Var}(\mathbf{y}_N) \end{bmatrix}$$

Canonical links for GLMS

Distribution	Canonical link function
Normal	$g(\mu) = \mu$
Gamma	$g(\mu) = \frac{1}{\mu}$
Poisson	$g(\mu) = \log(\mu)$
Bernoulli	$g(\mu) = \log\left(\frac{\mu}{1 - \mu}\right)$

Fisher scoring with normal linear model and canonical link

GLM model $\mathbf{y}_i \sim \text{Normal}(\mu_i, \sigma^2)$

$$\eta_i = x_i^T \beta$$

$$\eta_i = \mu_i$$

Score $\mathbf{u}_\beta = \frac{1}{\sigma^2} X^T (\mathbf{y} - X\beta)$

Fisher information matrix $F_\beta = \frac{1}{\sigma^2} X^T X$

Iteratively re-weighted least squares update

$$\beta^{(t+1)} = \beta^{(t)} + (X^T X)^{-1} X^T (\mathbf{y} - X\beta^{(t)})$$

Fisher scoring with normal linear model and canonical link

Let's simplify

$$\begin{aligned}\beta^{(t+1)} &= \beta^{(t)} + (X^T X)^{-1} X^T (\mathbf{y} - X\beta^{(t)}) \\ &= \beta^{(t)} + (X^T X)^{-1} X^T \mathbf{y} - (X^T X)^{-1} X^T X \beta^{(t)} \\ &= (X^T X)^{-1} X^T \mathbf{y}\end{aligned}$$

The update no longer depends on the previous iterate — Fisher scoring converges in one single step to the exact **ordinary least squares solution**

Estimating $\text{Var}(\mathbf{y})$

The outcome variance in a generalized linear model is determined by the distribution and its variance function

$$\text{Var}(\mathbf{y}_i) = a(\phi)b''(\theta) = \phi V(\mu_i)$$

Variance function 

	ϕ	$V(\mu_i)$
Normal	σ^2	1
Gamma	ϕ	μ_i^2
Poisson	1	μ_i
Bernoulli	1	$\mu_i(1 - \mu_i)$

Variance function describes how the variance changes with the mean

Estimating $\text{Var}(\mathbf{y})$

After our iterative algorithm has converged, we have $\hat{\mu}_i = g^{-1}(x_i^T \hat{\beta})$

	ϕ	$V(\mu_i)$
Normal	σ^2	1
Gamma	ϕ	μ_i^2
Poisson	1	μ_i
Bernoulli	1	$\mu_i(1 - \mu_i)$



We only really need to solve for ϕ in $\text{Var}(\mathbf{y}_i) = a(\phi)b''(\theta) = \phi V(\mu_i)$

Estimating the dispersion

Two ways - Pearson estimator or Deviance-based estimator of dispersion

Pearson estimator $\hat{\phi} = \frac{1}{N - p} \sum_{i=1}^N \frac{(\mathbf{y}_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$

Unbiased $E \left(\sum_{i=1}^N \frac{(\mathbf{y}_i - \mu_i)^2}{V(\mu_i)} \right) = \phi$

Consistent $\hat{\phi} \rightarrow \phi \text{ as } N \rightarrow \infty$

Asymptotically normal $\frac{\mathbf{y}_i - \mu_i}{\sqrt{V(\mu_i)}} \stackrel{\text{approx}}{\sim} \text{Normal}(0, \phi)$

Estimating the dispersion

Pearson estimator $\hat{\phi} = \frac{1}{N-p} \sum_{i=1}^N \frac{(\mathbf{y}_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$

Asymptotically normal $\frac{\mathbf{y}_i - \mu_i}{\sqrt{V(\mu_i)}} \underset{\sim}{\text{approx}} \text{Normal}(0, \phi)$

$$\frac{\mathbf{y}_i - \mu_i}{\sqrt{\phi V(\mu_i)}} \underset{\sim}{\text{approx}} \text{Normal}(0, 1)$$

The sum of squared standard normal rvs has the Chi-squared distribution

$$\sum_{i=1}^N \left(\frac{\mathbf{y}_i - \mu_i}{\sqrt{\phi V(\mu_i)}} \right)^2 = \frac{1}{\phi} \sum_{i=1}^N \frac{(\mathbf{y}_i - \mu_i)^2}{V(\mu_i)} = \frac{1}{\phi} X^2 \underset{\sim}{\text{approx}} \chi_{(N-p)}^2$$

Estimating the dispersion

Pearson estimator $\hat{\phi} = \frac{1}{N-p} \sum_{i=1}^N \frac{(\mathbf{y}_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}$

The sum of squared standard normal rvs has the Chi-squared distribution

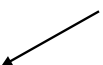
$$\sum_{i=1}^N \left(\frac{\mathbf{y}_i - \mu_i}{\sqrt{\phi V(\mu_i)}} \right)^2 = \frac{1}{\phi} \sum_{i=1}^N \frac{(\mathbf{y}_i - \mu_i)^2}{V(\mu_i)} = \frac{1}{\phi} X^2 \stackrel{\text{approx}}{\sim} \chi_{(N-p)}^2$$

$$\mathbb{E} \left(\frac{X^2}{\phi} \right) = N - p \quad (\text{the mean of chi-squared distribution})$$

$$\mathbb{E}(\hat{\phi}) = \frac{\mathbb{E}(X^2)}{N-p} = \phi \quad (\text{Pearson is an unbiased estimator})$$

In practice, we first calculate $\hat{\mu}_i$ then find $\hat{\phi}$ and finally $\widehat{\text{Var}}(\mathbf{y}_i) = \hat{\phi} V(\hat{\mu}_i)$

Estimating the dispersion

Deviance-based estimator $\hat{\phi} = \frac{D}{N - p}$  Residual deviance of the fitted GLM

Asymptotically equivalent to Pearson

Residual deviance $D(y, \hat{\mu}) = 2 [l(y, y) - l(y, \hat{\mu})]$

$$\text{where } l(y, \mu) = \sum_{i=1}^N \frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{a(\phi)} + c(y_i, \phi)$$

$l(y, y)$ is called **the saturated model** - every observation gets its own parameter and the fitted mean equal the observed outcome

The best possible log-likelihood any model could achieve on the data

Estimating the dispersion

Unit deviance (the contribution of a single observation, note that its > 0)

$$d_i = 2 [l(y_i, y_i) - l(y_i, \hat{\mu}_i)], \text{ where } l(y_i, \mu_i) = \frac{y_i \theta(\mu_i) - b(\theta(\mu_i))}{a(\phi)} + c(y_i, \phi)$$

Deviance residual $r_{D,i} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i}$

$\text{sign}(y_i - \hat{\mu}_i)$ indicates whether the fitted mean under- or over-estimates the observation

$$\text{sign}(x) = \begin{cases} 1, & \text{if } x \geq 0 \\ -1, & \text{if } x < 0 \end{cases}$$

The residual deviance $\sum_{i=1}^N r_{D,i}^2 = D$ and $\frac{1}{\phi} D \approx \chi_{(N-p)}^2$

Estimating the dispersion

Because

$$\sum_{i=1}^N r_{D,i}^2 = D, \quad r_{D,i} = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i} \quad \text{and} \quad \frac{1}{\phi} D \approx \chi_{(N-p)}^2$$

$$\mathbb{E} \left(\frac{D}{N-p} \right) = \phi$$

The deviance based estimator $\hat{\phi} = \frac{D}{N-p}$

Asymptotically unbiased

Consistent

Asymptotically normal estimator of the dispersion

Deviance based estimator is the default in R function glm

Confidence intervals for the link function

The maximum likelihood estimator for the link function $g(\mu) = \eta = x^T \beta$ is

$$\widehat{g(\mu)} = \hat{\eta} = x^T \hat{\beta}$$

and the estimated variance for $\hat{\eta} = x^T \hat{\beta}$ is

$$\text{Var}(x^T \hat{\beta}) = x^T \widehat{\text{Cov}}(\hat{\beta}) x = x^T (X^T \widehat{W} X)^{-1} x$$

and approximately

$$x^T \hat{\beta} \sim \text{Normal}(x^T \beta, x^T (X^T \widehat{W} X)^{-1} x)$$

Therefore approximate $100(1 - \alpha)\%$ CI for $\eta = x^T \beta$

$1 - \alpha/2$ quantile of
standard normal
distribution

$$[x^T \hat{\beta} - z_{\alpha/2} \sqrt{x^T \widehat{\text{Cov}}(\hat{\beta}) x}, x^T \hat{\beta} + z_{\alpha/2} \sqrt{x^T \widehat{\text{Cov}}(\hat{\beta}) x}]$$

Confidence intervals for the outcome mean

Confidence interval for the link function is

$$[x^T \hat{\beta} - z_{\alpha/2} \sqrt{x^T \widehat{\text{Cov}}(\hat{\beta}) x}, x^T \hat{\beta} + z_{\alpha/2} \sqrt{x^T \widehat{\text{Cov}}(\hat{\beta}) x}] = [\eta_{\alpha/2}^L, \eta_{\alpha/2}^U]$$

And using the inverse of the monotonically increasing $g(\mu)$
we obtain the confidence interval for the outcome mean

$$\left[g^{-1}(\eta_{\alpha/2}^L), g^{-1}(\eta_{\alpha/2}^U) \right]$$

What are the outcome confidence intervals used for?

Understanding the precision of estimated effects

They show how tightly the model estimates the mean at a given covariate pattern

Comparing expected responses between groups

If CIs for μ overlap or not, you can assess differences in mean levels (not outcome variability)

Plotting fitted curves with confidence bands

Model interpretation (not prediction/forecasting)

They help describe the model's conclusions about the underlying relationship—but they do not show uncertainty in actual future outcomes

Prediction intervals

Prediction intervals tell us the how a completely **new** observation would probabilistically behave given our model fit

Note the difference to confidence intervals that quantify the quality of the current fit of the selected model

There are analytical approximations on how to calculate the prediction intervals

When the distribution of the observation is “simple”

The most general method of investigating the prediction intervals is to use simulation method

Simulation method for prediction intervals

Algorithmic outline for generating the prediction intervals at x_0 :

1. Draw a random β^* from the asymptotic normal distribution

$$\beta^* \sim \text{Normal}(\hat{\beta}, \widehat{\text{Cov}}(\hat{\beta}))$$

2. Calculate $\mu^* = g^{-1}(x_0^T \beta^*)$

3. Draw simulated outcomes $y^* \sim f(\mu^*, \hat{\phi})$

4. Repeat 1.-3. many times

5. Use quantiles of simulated y^* as the prediction interval

Example: Logistic-regression

Let's simulate data from logistic regression model

$$\mathbf{y}_i \sim \text{Ber}(\mu_i) = f(y_i \mid \mu_i) = \mu^{y_i} (1 - \mu_i)^{1-y_i}$$

$$\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$$

$$\mu_i = g^{-1}(\eta_i) = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \mu_i(1 - \mu_i)$$

Example: Poisson-regression

Let's simulate data from Poisson regression model

$$\mathbf{y}_i \sim \text{Poisson}(\mu_i) = f(y_i \mid \mu_i) = \frac{\mu_i^{y_i} \exp(-\mu_i)}{y_i!}$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \exp(\eta_i)$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \mu_i$$

Example: Normal linear model

Let's simulate data from Normal linear model

$$\mathbf{y}_i \sim \text{Normal}(\mu_i, \sigma^2) = f(y_i \mid \mu_i) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{1}{2} \frac{(y_i - \mu_i)^2}{\sigma^2}\right)$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \eta_i$$

$$\mathbb{E}(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \sigma^2$$

Example: Gamma-regression

Let's simulate data from Gamma regression model

$$y_i \sim \text{Gamma}(\mu_i, \phi) = f(y_i \mid \mu_i, \phi) = \frac{1}{\Gamma(\frac{1}{\phi})} \left(\frac{1}{\phi \mu_i} \right)^{\frac{1}{\phi}} y_i^{\frac{1}{\phi}-1} \exp \left(-\frac{y_i}{\phi \mu_i} \right)$$

$$\eta_i = x_i^T \beta$$

$$\mu_i = g^{-1}(\eta_i) = \exp(\eta_i)$$

$$E(\mathbf{y}_i) = \mu_i$$

$$\text{Var}(\mathbf{y}_i) = \phi \mu_i^2$$

Compare the common gamma distribution formulation (used in R) and the mean & dispersion - formulation above

Common:

$$f(y_i \mid \alpha, \theta_i) = \frac{1}{\Gamma(\alpha) \theta_i^\alpha} y_i^{\alpha-1} \exp \left(-\frac{y_i}{\theta_i} \right)$$

where

$$\alpha = \frac{1}{\phi}$$

$$\theta_i = \mu_i \phi$$

End of Lecture6